Pixels, Frames & Transformers

Images and Words: CLIP · Module 09

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Learning from Captions Instead of Labels

CLIP (Radford et al., 2021) learned from about 400 million images with the text that came with them on the web.

- An image encoder (a ViT) and a text encoder
- In each batch, every image must pick out its own caption among all captions, and every caption its own image
- **Contrastive** learning: pull matching pairs together, push the rest apart



the diagonal should score high, everything else low

One Space for Images and Sentences

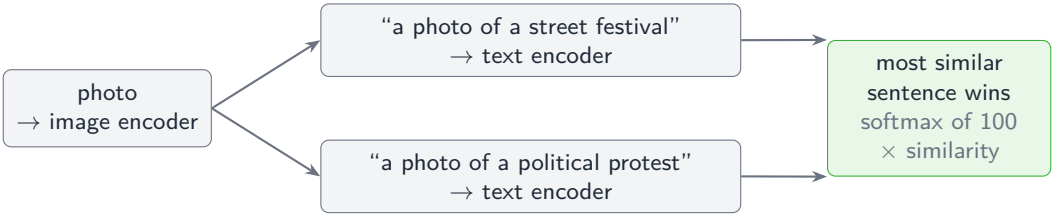
Both encoders map into the same space (512 numbers for CLIP ViT-B/32). Normalize every vector to length 1, and the dot product of an image and a sentence is their **cosine similarity**.

This lets you:

- classify with words (**zero-shot**)
- search images with a sentence
- define a measurement scale with two sentences

Similarities are relative. Values usually fall between about 0.15 and 0.35, and a few hundredths matter. Compare one image with several sentences, or one sentence with several images. A similarity of 0.27 does not mean "27% protest".

Zero-Shot Classification



No labeled examples needed: you describe each category in words. Every choice of words is a measurement decision, so it needs the same scrutiny as a codebook definition.

The Wording Changes the Result

- A few changed words can move accuracy by ten points or more, and the share labeled "protest" even more
- Negation fails: "a photo that does not show a protest" mostly registers *protest*
- **Prompt ensembles** average several templates ("a photo of . . . ", "a news photo of . . . ") so no single wording dominates

Choose the wording on data you don't score on
Picking the best prompt on the test set is tuning on the test set. Compare wordings on training images (or a separate validation sample), then score once.

Report the spread across reasonable wordings, not just the winner.

Zero-Shot or Labels?

A linear probe on the same embeddings uses your labeled images. With a few hundred labels, it usually beats zero-shot by a clear margin.

Zero-shot is useful when

- you have no labels yet
- you need many categories quickly
- you want candidates to hand-code

SigLIP 2 (Google, 2025, Apache 2.0)
A newer image-text model. It scores each image-caption pair on its own instead of against the whole batch, and it is usually stronger than the original CLIP. Same workflow, different `model_name`.

A prompt chosen for one model is not necessarily best for another.

Searching a Collection with Words

Embed every image once; then any sentence is a query.

- "packaging with a cartoon character"
- "a chocolate cereal for children"

Image + words: add the embedding of a phrase to the embedding of an image ("this box, *with a cartoon character*") to find similar items that differ in one respect.

Good for
getting into a large collection, finding examples and counterexamples for a codebook, and building hand-coding samples that are rich in rare cases.

Search results are suggestions to look at, not measurements.

A Measurement Scale Defined in Words

Score each image by its similarity to one pole minus its similarity to the other:

"a wholesome, natural breakfast cereal with whole grains"

versus

"a sweet, sugary breakfast cereal for children, a fun treat"

Like the poles in the embedding playground of Bags, Vectors & Transformers, but for images.

Validate it
Against human ratings of the concept you mean. Nutri-Score measures something related but different: what is *in* the box, not what the box *suggests*. Mismatches can be the finding.

Check the wording: rebuild the scale with other poles and see how much items move.

What CLIP Learned About People

CLIP learned from web captions, with their habits about who does what. Its own authors found that some groups in FairFace were more often put into crime-related or non-human categories (Radford et al., 2021, section 7).

An audit with a known answer: every Congress portrait shows a politician. How often does CLIP pick "a politician" from a list of occupations, for women and for men?

The label set shapes the result
Add "a member of Congress" or remove "a nurse" and "a secretary", and both the share correct and the gap between women and men change.

Report the full label set, and check that conclusions hold across reasonable alternatives (compare Schwemmer et al., 2020, in module 07).

In the Notebook

- Images and sentences in one space
- Zero-shot classification of your 40 coded images, with prompts chosen on training data
- Zero-shot versus a linear probe, and SigLIP 2 as an alternative
- Searching the packaging with words, and with an image plus words
- A “healthy-looking” scale, checked against Nutri-Score
- An occupation audit on the Congress portraits

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